SUJANA ACHARYA

AI/ML Research Engineer | AI Safety, Agentic Systems & Applied AI

acharyasujana17@gmail.com | Lalitpur, Nepal | github.com/sujanaacharya | linkedin.com/in/sujanaacharya17

Research Profile

AI/ML research engineer with more than two years of industry experience at a Korean EdTech company, spanning LLMs, automated assessment, speech AI, retrieval, multimodal systems, evaluation, and production ML infrastructure. Current research centers on **AI safety**: authorization, autonomous-agent behavior, deterministic safeguards, and evidence-grounded vulnerability analysis. Ongoing projects include **CHANAKYA** for Web2 broken-access-control/IDOR research and **SafetyBox** for autonomous-agent authorization and control. Long-term interests include **AI safety and governance, advanced autonomous systems, and AGI**.

Research Interests

- **AI Safety:** authorization, access-control failures, agent boundary enforcement, deterministic verification, evaluation of unsafe autonomous behavior, evidence-grounded AI reasoning.
- **AI Governance:** technical governance mechanisms, accountability, controllability, deployment safeguards, auditing, and evaluation standards for increasingly autonomous AI systems.
- **Agentic AI & AGI:** autonomous agents, long-horizon reasoning, tool use, capability control, failure modes, alignment, robustness, and system-level safety.
- **Computational Neuroscience & NeuroAI:** biologically inspired learning and evaluation, neural correlates of cognition and judgment, and models that connect artificial and biological intelligence.
- **AI in Healthcare:** medical visual question answering, medical imaging and vision-language models, and safe, reliable, evidence-grounded AI for clinical settings.
- **Applied AI Research:** LLM evaluation, retrieval-augmented generation, multimodal systems, speech AI, automated scoring, and production inference.

Ongoing AI Safety Research

CHANAKYA: Web2 Vulnerability Detection and Authorization Research *Ongoing*

- Building a research-oriented vulnerability-detection system for source-code analysis, with a primary focus on **IDOR/BOLA, missing authorization, broken access control, and related authorization failures**.
- Developing a hybrid architecture that combines deterministic authorization evidence with bounded AI investigation, so model conclusions can be traced back to concrete code and control-flow evidence rather than unsupported guesses.
- Investigating repository navigation, framework-aware code understanding, authorization evidence extraction, inter-procedural reasoning, and controlled AI-assisted vulnerability analysis.
- Current work emphasizes Java/Spring-style authorization patterns while developing techniques that can generalize to broader Web2 application stacks.

SafetyBox: Autonomous-Agent Safety and Authorization *Ongoing*

- Researching how autonomous agents can exhibit unsafe, unintended, or "rogue" behavior when probabilistic reasoning is allowed to directly influence privileged actions.
- Exploring **deterministic task-scoped authorization, runtime mediation, capability restrictions, and effect-level enforcement** as mechanisms for bounding agent behavior.
- Studying how interpretation and reasoning can remain probabilistic while final authority and execution are constrained by deterministic checks.
- Developing adversarial and counterfactual authorization cases to analyze false-allow and false-deny behavior and to evaluate whether an agent can exceed its intended permissions.

Professional Experience

AI/ML Engineer **Oct 2026 - Present**

Smart Data Solutions, US-based company

AI/ML Engineer **Aug 2024 - Sep 2026**

NsDevil, Korean EdTech / AI Systems *Lalitpur, Nepal*

- Designed, built, evaluated, and deployed production AI systems spanning LLMs, speech processing, educational assessment, retrieval, multimodal AI, and personalized learning.
- Deployed and optimized transformer and LLM inference services using **PyTorch, Hugging Face, vLLM, FastAPI, Docker, CUDA, and GPU infrastructure**.
- Built automated scoring and evaluation pipelines for Korean-language educational systems, including rubric-based LLM scoring, model-comparison workflows, dataset analysis, and **Quadratic Weighted Kappa (QWK)** evaluation.
- Worked with Korean-language LLM systems including HyperCLOVA X and other transformer-based services; contributed to prompt design, model serving, scoring calibration, inference optimization, and production integration.
- Built speech-evaluation pipelines using Whisper-style ASR, forced alignment, pronunciation analysis, WER/CER/PER-style metrics, and downstream LLM-based semantic and fluency assessment.
- Developed retrieval and semantic-search systems using dense embeddings, FAISS/vector retrieval, reranking, database-backed embedding stores, and RAG.

- Contributed to production engineering and MLOps using **GitHub Actions, Docker/Compose, Nginx, Linux, PostgreSQL, CI/CD, GPU servers, and service monitoring**.
- Collaborated on research-to-production workflows where model quality had to be validated against human labels, rubric constraints, and measurable evaluation metrics before deployment.

Selected systems and projects:

- **QRMAC / Rubric Templating:** AI-assisted rubric and question-generation systems aligned to Korean curriculum standards, with structured evaluation, rule-backed rubrics, and production LLM integration.
- **Essay_V2:** automated essay-scoring architecture combining multiple LLMs, rubric constraints, category-level scoring, and quantitative agreement analysis.
- **Popkor:** Korean speech-evaluation pipeline combining ASR, forced alignment, pronunciation metrics, deterministic scoring signals, and LLM-based semantic/grammar/fluency evaluation.
- **AlphaHowl / LLM services:** production model APIs and LLM-serving infrastructure for internal AI applications.
- **AI Tutor / MathPad and related EdTech systems:** applied AI components for educational interaction, scoring, feedback, content generation, and multimodal learning workflows.

AI Research Intern **Apr 2024 - Jul 2024**
NsDevil *Lalitpur, Nepal*

- Built a retrieval-augmented generation system for Korean curriculum content using GPT APIs, FAISS, and embedding-based semantic retrieval.
- Analyzed student-performance data using classical machine-learning approaches and explored transformer, computer-vision, and vision-language architectures.
- Contributed to early-stage applied research that transitioned into broader production AI/ML engineering work.

Publications

B. Dahal, S. Bhatta, A. Shrestha, S. Acharya, S. Shakya. "MediVQA: Medical Visual Question Answering using VisualBERT." *European Journal of Applied Science, Engineering and Technology*, Vol. 3(5), pp. 96-104, 2025.
Role: Co-author (4th author).
DOI: https://doi.org/10.59324/ejaset.2025.3(5).08
Article: https://ejaset.com/index.php/journal/article/view/350

B. Dahal, S. Bhatta, S. Acharya, A. Shrestha, P. Acharya. "Automated Retail Billing: Streamlining Checkout with QR Codes and Object Tracking Using YOLOv8 and DeepSORT." *International Journal of Science, Engineering and Technology*, Vol. 12, Issue 5, 2024.
Role: Co-author (3rd author).
DOI: https://doi.org/10.61463/ijset.vol.12.issue5.253
Article: https://www.ijset.in/automated-retail-billing-streamlining-checkout-qr-codes-object-tracking-using-yolov8-deepsort/

Selected Technical Expertise

AI Safety & Agents	Authorization, IDOR/BOLA reasoning, access-control analysis, deterministic verification, autonomous-agent safeguards, capability boundaries, adversarial evaluation
LLM / NLP	LLM inference, prompt and system design, fine-tuning concepts, RAG, embeddings, reranking, BERT-family models, HyperCLOVA X, Mistral-class models, seq2seq, evaluation
ML / Deep Learning	PyTorch, Hugging Face Transformers/Datasets/Accelerate, scikit-learn, XGBoost, model evaluation, dataset analysis
Speech / Multimodal	Whisper-style ASR, forced alignment, pronunciation evaluation, OpenCV, YOLOv8, DeepSORT, ResNet, visual-language architectures
Evaluation	QWK, accuracy, precision/recall concepts, BLEU, ROUGE, WER/CER/PER-style metrics, benchmarking, human-vs-model agreement analysis
Programming	Python, C, C++, SQL
Backend / MLOps	FastAPI, Django REST Framework, Docker, Docker Compose, GitHub Actions, Linux, Nginx, CUDA, PostgreSQL, MySQL, SQLite, FAISS, GPU deployment

Education

Bachelor of Electronics, Communication and Information Engineering (BEI) **2019 - 2024**
Thapathali Engineering Campus, Institute of Engineering, Tribhuvan University *Kathmandu, Nepal*

- Foundations in artificial intelligence, algorithms and data structures, probability and statistics, numerical methods, operating systems, computer networks, signal processing, embedded systems, and engineering mathematics.

Research Training, Programs & Languages

- **Effective Altruism Introductory Programme:** completed September 2026; strengthened interest in AI safety, governance, long-term technological risk, and responsible decision-making around advanced AI.
- **IELTS Academic:** Overall Band 8.0.
- **Languages:** Nepali (native), English (professional working proficiency).